# AI Image Generator

## 8. Database

Version 2.1
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, Qwen-Image, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

# 8. Database

## 8.1 Overview

File: generations.db in the project folder. Engine: SQLite 3 through Python's standard library. One table, generations, one row per image produced by the web app. An index on created_at supports newest-first listing. The schema is created on first connection; columns added in later increments are listed in NEW_COLUMNS and added with ALTER TABLE when absent. Since v2.0 rows can be amended after creation: the Upscale bar writes the upscale columns onto an existing row with db.update().

## 8.2 Table: generations

### Identity and files

| Column | Type | Meaning |
|---|---|---|
| id | INTEGER PK | Auto-increment row id, shown as "#N" in the UI |
| created_at | TEXT | ISO-8601 local timestamp, seconds precision |
| image_path | TEXT | Absolute path of the native PNG |
| file_size_kb | REAL | Size of the native PNG |

### Prompt

| Column | Type | Meaning |
|---|---|---|
| user_prompt | TEXT | Exactly what was typed, including model and LoRA tags |
| final_prompt | TEXT | What was sent to the model after directive removal and style suffixes |
| negative_prompt | TEXT | Negative prompt used; empty for FLUX |
| parser_notes | TEXT | Parser decisions, auto-model note, LoRA notes and the sampler, joined with " - " |

### Settings

| Column | Type | Meaning |
|---|---|---|
| model_id | TEXT | Model source: Hugging Face repo id or local checkpoint path |
| width, height | INTEGER | Native generation size |
| steps | INTEGER | Denoising steps actually used (after model defaults) |
| guidance | REAL | Guidance scale actually used |
| seed | INTEGER | Seed used for this image |
| style | TEXT | Name of the matched style preset, or NULL |
| batch_index, batch_size | INTEGER | Position within a multi-image request |

### Timing

| Column | Type | Meaning |
|---|---|---|
| inference_seconds | REAL | Pipeline call, from start to CUDA synchronise |
| save_seconds | REAL | PNG encode and write of the native image |
| total_seconds | REAL | inference + save (excludes upscaling) |
| steps_per_second | REAL | steps / inference_seconds |

### GPU and software

| Column | Type | Meaning |
|---|---|---|
| gpu_name | TEXT | From torch device properties |
| vram_total_gb | REAL | Card memory |
| vram_peak_gb | REAL | Peak torch allocation during this image, including an in-prompt upscale |
| vram_reserved_gb | REAL | Allocator reservation after the image |
| compute_capability | TEXT | e.g. "8.6" |
| driver_version | TEXT | From nvidia-smi |
| cuda_version, torch_version, diffusers_version | TEXT | Library versions |

### Telemetry (from nvidia-smi)

| Column | Type | Meaning |
|---|---|---|
| temp_before_c, temp_after_c | REAL | GPU temperature before the pipeline call and right after it |
| power_before_w, power_after_w | REAL | Board power draw at the same two moments |
| power_limit_w | REAL | Configured power limit |
| gpu_util_after_pct | REAL | GPU utilisation at the after snapshot |
| sm_clock_mhz, mem_clock_mhz | REAL | Clock speeds at the after snapshot |
| fan_after_pct | REAL | Fan speed |
| pstate | TEXT | Performance state, P0 to P8 |

### Upscaling (written at generation or later by the Upscale bar)

| Column | Type | Meaning |
|---|---|---|
| upscale_target | INTEGER | Requested long side in pixels |
| upscaled_width, upscaled_height | INTEGER | Final size after resample and crop |
| upscaled_path | TEXT | Absolute path of the upscaled PNG |
| upscaled_size_kb | REAL | Size of that file |
| upscale_seconds | REAL | Time of the upscaling stage |
| upscale_passes | INTEGER | Number of x4 passes (1 or 2) |
| upscaler | TEXT | Architecture and weight file name |

### Catch-all

| Column | Type | Meaning |
|---|---|---|
| extra_json | TEXT | JSON of any record keys that have no column, for forward compatibility |

## 8.3 Example queries

Average inference time by model:

```
SELECT model_id, COUNT(*) AS n, ROUND(AVG(inference_seconds), 2) AS avg_s,
       ROUND(AVG(steps_per_second), 2) AS steps_s
FROM generations GROUP BY model_id ORDER BY n DESC;
```

Hottest runs:

```
SELECT id, temp_after_c, power_after_w, user_prompt
FROM generations ORDER BY temp_after_c DESC LIMIT 5;
```

Every 4K or larger image with its file:

```
SELECT id, upscaled_width, upscaled_height, upscale_seconds, upscaled_path
FROM generations WHERE upscale_target >= 3840;
```

Estimated GPU energy of all generations, as the Admin panel computes it:

```
SELECT ROUND(SUM(inference_seconds * (power_before_w + power_after_w) / 2) / 3600, 1) AS wh
FROM generations;
```

## 8.4 Access from Python

```
import db
rows = db.fetch(limit=10, search="cabin")
one = db.get(4)
db.update(4, {"upscaled_width": 3840})
totals = db.stats()
```